

7.1 Using equation 2.249:

$$p(\mathbf{x}|\mathcal{C}_1) = \frac{1}{N_1} \sum_{\mathbf{x}_n \in \mathcal{C}_1} \frac{1}{h^D} k(\mathbf{x}, \mathbf{x}_n)$$

and

$$p(\mathbf{x}|\mathcal{C}_2) = \frac{1}{N_2} \sum_{\mathbf{x}_n \in \mathcal{C}_2} \frac{1}{h^D} k(\mathbf{x}, \mathbf{x}_n)$$

Since the two classes have equal prior probability,

$$p(\mathcal{C}_1) = p(\mathcal{C}_2) = 1/2$$

$$p(\mathbf{x}, \mathcal{C}_1) = p(\mathbf{x}|\mathcal{C}_1) p(\mathcal{C}_1) = \frac{1}{2N_1} \sum_{n \in \mathcal{C}_1} \frac{1}{h^D} k(\mathbf{x}, \mathbf{x}_n)$$

$$p(\mathbf{x}, \mathcal{C}_2) = p(\mathbf{x}|\mathcal{C}_2) p(\mathcal{C}_2) = \frac{1}{2N_2} \sum_{n \in \mathcal{C}_2} \frac{1}{h^D} k(\mathbf{x}, \mathbf{x}_n)$$

$$p(\mathbf{x}) = p(\mathbf{x}, \mathcal{C}_1) + p(\mathbf{x}, \mathcal{C}_2)$$

$$\begin{aligned} p(\mathcal{C}_1|\mathbf{x}) &= \frac{p(\mathbf{x}, \mathcal{C}_1)}{p(\mathbf{x})} \\ &= \frac{\frac{1}{2N_1} \sum_{n \in \mathcal{C}_1} \frac{1}{h^D} k(\mathbf{x}, \mathbf{x}_n)}{\frac{1}{2N_1} \sum_{n \in \mathcal{C}_1} \frac{1}{h^D} k(\mathbf{x}, \mathbf{x}_n) + \frac{1}{2N_2} \sum_{n \in \mathcal{C}_2} \frac{1}{h^D} k(\mathbf{x}, \mathbf{x}_n)} \\ &= \frac{\frac{1}{N_1} \sum_{n \in \mathcal{C}_1} k(\mathbf{x}, \mathbf{x}_n)}{\frac{1}{N_1} \sum_{n \in \mathcal{C}_1} k(\mathbf{x}, \mathbf{x}_n) + \frac{1}{N_2} \sum_{n \in \mathcal{C}_2} k(\mathbf{x}, \mathbf{x}_n)} \end{aligned}$$

So, for any $\mathbf{x}'$, we can minimize classification errors if $\mathbf{x}'$ is classified as belonging to $\mathcal{C}_1$ when

$$\begin{aligned} &\frac{\frac{1}{N_1} \sum_{n \in \mathcal{C}_1} k(\mathbf{x}', \mathbf{x}_n)}{\frac{1}{N_1} \sum_{n \in \mathcal{C}_1} k(\mathbf{x}', \mathbf{x}_n) + \frac{1}{N_2} \sum_{n \in \mathcal{C}_2} k(\mathbf{x}', \mathbf{x}_n)} \\ &> \\ &\frac{\frac{1}{N_2} \sum_{n \in \mathcal{C}_2} k(\mathbf{x}', \mathbf{x}_n)}{\frac{1}{N_1} \sum_{n \in \mathcal{C}_1} k(\mathbf{x}', \mathbf{x}_n) + \frac{1}{N_2} \sum_{n \in \mathcal{C}_2} k(\mathbf{x}', \mathbf{x}_n)} \end{aligned}$$

and vice versa.

The denominators being the same, this condition becomes:

$$\frac{1}{N_1} \sum_{n \in \mathcal{C}_1} k(\mathbf{x}', \mathbf{x}_n) > \frac{1}{N_2} \sum_{n \in \mathcal{C}_2} k(\mathbf{x}', \mathbf{x}_n)$$

If the kernel is a simple dot product, then this becomes:

$$\begin{aligned} \frac{1}{N_1} \sum_{n \in \mathcal{C}_1} \mathbf{x}'^T \mathbf{x}_n &> \frac{1}{N_2} \sum_{n \in \mathcal{C}_2} \mathbf{x}'^T \mathbf{x}_n \\ \frac{1}{N_1} \cdot \mathbf{x}'^T \left(\sum_{n \in \mathcal{C}_1} \mathbf{x}_n \right) &> \frac{1}{N_2} \cdot \mathbf{x}'^T \left(\sum_{n \in \mathcal{C}_2} \mathbf{x}_n \right) \end{aligned}$$

$\sum_{n \in \mathcal{C}_1} \mathbf{x}_n$ is just the sum of all points in $\mathcal{C}_1$, which is the same as N_1 times the mean of all the points in $\mathcal{C}_1$. Same logic applies to $\mathcal{C}_2$. So if we let the means be represented with $\bar{\mathbf{x}}_1$ and $\bar{\mathbf{x}}_2$, the condition becomes:

$$\begin{aligned} \frac{1}{N_1} \mathbf{x}'^T (N_1 \cdot \bar{\mathbf{x}}_1) &> \frac{1}{N_2} \mathbf{x}'^T (N_2 \cdot \bar{\mathbf{x}}_2) \\ \mathbf{x}'^T \bar{\mathbf{x}}_1 &> \mathbf{x}'^T \bar{\mathbf{x}}_2 \end{aligned}$$

Finally, if the kernel is of the form $k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x})^T \phi(\mathbf{x}')$, the condition becomes:

$$\begin{aligned} \frac{1}{N_1} \sum_{n \in \mathcal{C}_1} \phi(\mathbf{x}')^T \phi(\mathbf{x}_n) &> \frac{1}{N_2} \sum_{n \in \mathcal{C}_2} \phi(\mathbf{x}')^T \phi(\mathbf{x}_n) \\ \phi(\mathbf{x}')^T \left(\frac{1}{N_1} \sum_{n \in \mathcal{C}_1} \phi(\mathbf{x}_n) \right) &> \phi(\mathbf{x}')^T \left(\frac{1}{N_2} \sum_{n \in \mathcal{C}_2} \phi(\mathbf{x}_n) \right) \\ \phi(\mathbf{x}')^T \bar{\phi}_1 &> \phi(\mathbf{x}')^T \bar{\phi}_2 \end{aligned}$$